

$$i) \Pr(R_1 = x_1, R_2 = x_2, \dots) = \Pr(R_1 = x_1) \dots \Pr(R_n = x_n).$$

$$\Rightarrow \Pr(R_1 = x_1 | R_2 = x_2) = \Pr(R_1 = x_1),$$

$$\text{即 } \Pr(A|B) = \Pr(A)$$

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Then H_1 is independent of M , since

$$\Pr[M = 1] = 1/4 = \Pr[M = 1 | H_1 = 1] = \Pr[M = 1 | H_1 = 0]$$

$$\Pr[M = 0] = 3/4 = \Pr[M = 0 | H_1 = 1] = \Pr[M = 0 | H_1 = 0]$$

This example is an instance of:

Lemma 18.2.1. *Two events are independent iff their indicator variables are independent.*

The simple proof is left to Problem 18.1.

Intuitively, the independence of two random variables means that knowing some information about one variable doesn't provide any information about the other one. We can formalize what "some information" about a variable R is by defining it to be the value of some quantity that depends on R . This intuitive property of independence then simply means that functions of independent variables are also independent:

Lemma 18.2.2. *Let R and S be independent random variables, and f and g be functions such that $\text{domain}(f) = \text{codomain}(R)$ and $\text{domain}(g) = \text{codomain}(S)$. Then $f(R)$ and $g(S)$ are independent random variables.*

The proof is another simple exercise left to Problem 18.30.

As with events, the notion of independence generalizes to more than two random variables.

Definition 18.2.3. Random variables $R_1, R_2, \dots, R_n$ are mutually independent iff for all $x_1, x_2, \dots, x_n$, the n events

$$[R_1 = x_1], [R_2 = x_2], \dots, [R_n = x_n]$$

are mutually independent. They are k -way independent iff every subset of k of them are mutually independent.

Lemmas 18.2.1 and 18.2.2 both extend straightforwardly to k -way independent variables.

18.3 Distribution Functions 分布函数.

A random variable maps outcomes to values. The probability density function, $\text{PDF}_R(x)$, of a random variable, R , measures the probability that R takes the value